Numerical Nondegeneracy and Strict Hedgehog-Sector Stability in the Frozen DTFE Matter Branch

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Abstract

We study the frozen DTFE matter-sector hedgehog branch as a finite-energy nonlinear profile problem with Skyrme-type scaling balance. The aim is narrow and exact. We do not attempt a full-field stability theorem for the coupled parent theory, and we do not write a wider unification paper here. We isolate the hedgehog sector, determine the exact analytical obstruction to strict stability, and record the current nondegeneracy status at the correct theorem level.

In the monotone hedgehog class, existence of a finite-energy minimiser, strict monotonicity, smoothness on $(0, \infty)$, regular origin expansion, quadratic far-field decay, virial balance, and non-negativity of the hedgehog Jacobi bottom can all be established analytically. These results reduce strict positivity of the hedgehog Jacobi operator to one remaining issue: whether the regular Jacobi branch generated by differentiation of the local shooting family lands in the admissible decaying zero-mode class at infinity.

That reduction is the main exact theorem of the paper. The numerical contribution is then directed only at the remaining obstruction. A sharply localised final shooting window

$$a_* \in [2.007528219473,\ 2.007528246180]$$

is obtained for the monotone decaying branch. Using direct asymptotic coefficient extraction, finite-radius input bands at $R_0 = 100$, and exact A/B tail transport to infinity, the current proof chain yields the bracket-wide asymptotic band

$$A(\infty) \in [-2.980428418910939237 \times 10^{-1},\ -2.980390555581560186 \times 10^{-1}],$$

which lies strictly below zero.

This does not yet constitute the final interval-certified closure of the strict-stability theorem. What has been achieved is sharper. The analytical obstruction

has been reduced exactly to one nondegeneracy condition, and the current asymptotic proof chain resolves that condition numerically in a bracket-wide form that stays uniformly away from zero. The remaining task is therefore no longer discovery but certification: replace the present finite-radius floating-point inputs and sampled coefficient bounds by validated interval enclosures and a rigorous tail lemma. The paper is proof-first in structure, paper-second in scope, and exact about the current boundary.

1 Introduction

The present paper is about one problem only: strict hedgehog-sector stability for the frozen DTFE matter branch.

That scope matters. The frozen DTFE matter sector already supports a broader architectural interpretation inside the larger Darke Theory corpus, but none of that broader interpretive weight is needed here. The present task is not to restate that architecture, and not to inflate a local result into a larger claim before it has been earned. The task is to determine whether the frozen charge-one hedgehog branch is strictly stable within the hedgehog perturbation sector, and to record the exact theorem status of that question.

The problem has two layers.

First, there is an analytical layer. In the monotone hedgehog class one can isolate a finite-energy minimiser, derive the exact profile equation, prove regularity and asymptotic decay, establish the virial identity, compute the second variation, and show that the bottom of the hedgehog Jacobi spectrum is non-negative. Once that is done, strict positivity is reduced to one remaining obstruction: the possible existence of an admissible decaying Jacobi zero mode.

Secondly, there is a numerical layer. If the regular local shooting family is differentiated with respect to its origin slope parameter, the resulting branch solves the Jacobi equation. Strict positivity is therefore equivalent to showing that this regular Jacobi branch does not land in the decaying class at infinity. The numerical task is not to find a new principle. It is to determine whether that nondegeneracy target is in fact satisfied.

The force of the current position is therefore precise. Analytically, strict positivity has been reduced to a single transversality question. Numerically, that transversality question now carries a coherent end-to-end proof chain in the correct asymptotic variable. The role of the present paper is to state both parts exactly, with no theorem inflation and no artificial softening where genuine theorems already exist.

2 Scope and exact theorem status

The exact status of the result is now the following. The analytical reduction theorem is proved. The numerical nondegeneracy target is strongly and coherently resolved in the correct asymptotic variable. The only remaining unresolved step is formal certification of that asymptotic exclusion by validated interval methods.

1. **Analytically established.** Within the monotone hedgehog class, existence, strict monotonicity, smoothness on $(0, \infty)$, the regular origin expansion, quadratic far-field decay, virial balance, the exact Jacobi operator, endpoint asymptotics of the effective potential, non-negativity of the hedgehog Jacobi bottom, and the reduction of strict positivity to a single nondegeneracy condition are all established.

2. **Numerically established.** The monotone decaying branch is sharply localised in a narrow final shooting window. The regular Jacobi branch generated by the shooting derivative exhibits a stable nonzero finite-radius transversality signal, a stable nonzero asymptotic growing coefficient, and a bracket-wide transported asymptotic band which remains strictly negative throughout the final window.

3. **Certification not yet closed.** The asymptotic exclusion is presently numerical rather than interval-certified. The remaining formal step is to replace the current bracket-wide band for $A(\infty)$ by a validated enclosure on the final shooting window together with a rigorous tail transfer estimate.

This is the exact posture adopted throughout the paper. Where theorem exists, theorem is stated. Where only numerical support exists, that support is recorded as numerical support. Where the proof boundary remains open, that boundary is stated explicitly.

3 Frozen hedgehog energy and admissible class

We study the frozen hedgehog energy

$$E[f] = 4\pi \int_0^\infty \left(r^2 f'(r)^2 + 2\sin^2 f(r) + 2\sin^2 f(r)\, f'(r)^2 + \frac{\sin^4 f(r)}{r^2} \right) \mathrm{d}r \tag{1}$$

under the topological boundary conditions

$$f(0) = \pi, \qquad \lim_{r\to\infty} f(r) = 0. \tag{2}$$

Definition 3.1 (Monotone hedgehog admissible class)**.** The monotone admissible hedgehog

class is

$$\mathcal{A}_{\text{mon}} = \left\{ f : [0,\infty) \to \mathbb{R} \;\middle|\; \begin{array}{l} f \text{ is absolutely continuous on compact intervals,} \\ f(0) = \pi, \quad \lim_{r\to\infty} f(r) = 0, \\ f \text{ is nonincreasing,} \\ E[f] < \infty \end{array} \right\}.$$

The energy splits naturally into quadratic and quartic pieces,

$$E[f] = E_2[f] + E_4[f], \tag{3}$$

where

$$E_2[f] = 4\pi \int_0^\infty \left(r^2 f'(r)^2 + 2\sin^2 f(r) \right) \mathrm{d}r, \tag{4}$$

and

$$E_4[f] = 4\pi \int_0^\infty \left(2\sin^2 f(r)\, f'(r)^2 + \frac{\sin^4 f(r)}{r^2} \right) \mathrm{d}r. \tag{5}$$

Lemma 3.2. *The class $\mathcal{A}_{\text{mon}}$ is nonempty.*

Proof. The explicit profile

$$f_0(r) = 2\arctan\left(\frac{1}{r}\right)$$

is smooth on $(0,\infty)$, satisfies $f_0(0) = \pi$, $f_0(\infty) = 0$, is strictly decreasing, and gives finite energy in (1). Hence $\mathcal{A}_{\text{mon}} \neq \varnothing$. □

3.1 Scaling structure

For the rescaled profile family

$$f_\lambda(r) = f(\lambda r), \qquad \lambda > 0, \tag{6}$$

a direct change of variables gives

$$E_2[f_\lambda] = \lambda^{-1} E_2[f], \qquad E_4[f_\lambda] = \lambda E_4[f]. \tag{7}$$

Hence

$$E[f_\lambda] = \lambda^{-1} E_2[f] + \lambda E_4[f]. \tag{8}$$

This is the finite-scale balance already familiar from the frozen DTFE matter sector. It blocks both zero-scale collapse and unlimited dispersion for nontrivial stationary branches.

4 Existence, monotonicity, regularity, and asymptotics

4.1 Existence in the monotone class

Proposition 4.1 (Existence of a monotone hedgehog minimiser)**.** *There exists* $f_* \in \mathcal{A}_{\mathrm{mon}}$ *such that*

$$E[f_*] = \inf_{f \in \mathcal{A}_{\mathrm{mon}}} E[f].$$

Proof. Take a minimising sequence $\{f_n\} \subset \mathcal{A}_{\mathrm{mon}}$. Since all integrand terms in (1) are nonnegative, the sequence has uniformly bounded local energy on every compact interval. The quadratic sector controls f_n' in L^2_{loc} with the radial weight, while the $\sin^2 f_n$ term controls the tails. Monotonicity blocks oscillatory escape and stabilises the boundary class in the limit.

Passing to a subsequence, one obtains local weak compactness and pointwise convergence almost everywhere. Lower semicontinuity of the nonnegative integral terms on compact intervals, together with monotone convergence on the tails, yields a limit $f_* \in \mathcal{A}_{\mathrm{mon}}$ with $E[f_*] \leq \liminf E[f_n]$. Hence f_* is a minimiser. □

4.2 Profile equation

A direct Euler–Lagrange computation for (1) gives the exact profile ODE

$$\left(r^2 + 2\sin^2 f\right) f'' + 2rf' + 2\sin f \cos f\, f'^2 - 2\sin f \cos f - \frac{2\sin^3 f \cos f}{r^2} = 0. \tag{9}$$

4.3 Strict monotonicity

Proposition 4.2 (Strict monotonicity)**.** *The minimiser* f_* *satisfies*

$$f_*'(r) < 0 \qquad \textit{for all } r > 0.$$

Proof. We already know that f_* is nonincreasing. Suppose there exists $r_0 > 0$ such that $f_*'(r_0) = 0$.

Insert this into (9). At r_0 one obtains

$$\left(r_0^2 + 2\sin^2 f_*(r_0)\right) f_*''(r_0) = 2\sin f_*(r_0) \cos f_*(r_0) \left(1 + \frac{\sin^2 f_*(r_0)}{r_0^2}\right).$$

If $f_*(r_0) \neq \pi/2$, then the right-hand side is nonzero, so $f_*''(r_0) \neq 0$. Hence f_*' changes sign near r_0, contradicting monotonicity.

If $f_*(r_0) = \pi/2$ and $f_*'(r_0) = 0$, then the Cauchy data $(f_*(r_0), f_*'(r_0)) = (\pi/2, 0)$ force the constant solution $f \equiv \pi/2$ by local uniqueness for the quasilinear ODE (9). That contradicts the boundary conditions (2). Therefore no such r_0 exists. □

4.4 Regularity

Proposition 4.3 (Smoothness on $(0, \infty)$)**.** *The minimiser f_* is C^∞ on $(0, \infty)$.*

Proof. On every compact interval $[\delta, R] \subset (0, \infty)$ the coefficient

$$r^2 + 2\sin^2 f_*(r)$$

is strictly positive, so the profile equation is uniformly quasilinear elliptic in one dimension. Standard bootstrap therefore upgrades weak regularity to smoothness on $(0, \infty)$. □

4.5 Regular origin expansion

The regular local branch has an odd expansion

$$f(r; a) = \pi - ar + b(a)r^3 + c(a)r^5 + \mathcal{O}(r^7), \qquad r \downarrow 0, \tag{10}$$

where

$$b(a) = \frac{a^3(a^2 + 2)}{15(2a^2 + 1)}, \tag{11}$$

and

$$c(a) = \frac{a^5\Big(92a^8 + 316a^6 + 459a^4 + 330a^2 + 120\Big)}{3150(2a^2 + 1)^3(5a^2 + 3)}. \tag{12}$$

Differentiating with respect to the shooting parameter a gives

$$\xi(r; a) := \partial_a f(r; a) = -r + b'(a)r^3 + c'(a)r^5 + \mathcal{O}(r^7), \tag{13}$$

with

$$b'(a) = \frac{a^2(2a^4 + 3a^2 + 2)}{5(2a^2 + 1)^2}. \tag{14}$$

4.6 Far-field decay

Proposition 4.4 (Quadratic decay)**.** *The minimiser satisfies*

$$f_*(r) = \frac{C}{r^2} + o(r^{-2}) \qquad (r \to \infty)$$

for some constant $C > 0$.

Proof. Since $f_*(r) \to 0$, the profile equation linearises to

$$r^2 f'' + 2rf' - 2f = 0$$

at infinity. A power ansatz $f \sim r^\alpha$ gives the indicial equation

$$\alpha(\alpha - 1) + 2\alpha - 2 = 0,$$

that is,

$$\alpha^2 + \alpha - 2 = 0 \quad \Longrightarrow \quad \alpha = 1,\, -2.$$

The r^1 branch is excluded by finite energy, so only the r^{-2} branch remains. □

4.7 Virial balance

Proposition 4.5 (Virial identity)**.** *If f_* is stationary under scaling, then*

$$E_2[f_*] = E_4[f_*].$$

Proof. Differentiate (8) at $\lambda = 1$:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} E[f_\lambda]\bigg|_{\lambda=1} = -E_2[f_*] + E_4[f_*].$$

Stationarity under scaling forces this derivative to vanish. □

5 Second variation and the hedgehog Jacobi operator

Let f be a critical profile and consider a perturbation $f + \varepsilon\eta$. The second variation takes the form

$$\delta^2 E[f](\eta) = 4\pi \int_0^\infty \Big(2\Big(r^2 + 2\sin^2 f\Big)\eta'^2 + V_{\text{eff}}(r)\eta^2\Big)\, \mathrm{d}r, \tag{15}$$

where

$$V_{\text{eff}}(r) = 4\cos(2f)(1 + f'^2) + \frac{4\sin^2 f\,(3\cos^2 f - \sin^2 f)}{r^2} - 8\frac{\mathrm{d}}{\mathrm{d}r}\Big(\sin f \cos f\, f'\Big). \tag{16}$$

The corresponding Jacobi operator is

$$\mathcal{J}_f \eta = -2\frac{\mathrm{d}}{\mathrm{d}r}\Big((r^2 + 2\sin^2 f)\eta'\Big) + V_{\text{eff}}(r)\eta. \tag{17}$$

5.1 Endpoint behaviour of the effective potential

Proposition 5.1 (Endpoint asymptotics of V_{eff})**.** *For the regular branch with origin slope a,*

$$V_{\text{eff}}(r) = 4 + 8a^2 + \mathcal{O}(r^2) \qquad (r \downarrow 0),$$

and for the decaying branch,

$$V_{\text{eff}}(r) = 4 + \mathcal{O}(r^{-4}) \qquad (r \to \infty).$$

Proof. Insert the regular origin expansion (10) into (16). The first nontrivial terms produce

$$V_{\text{eff}}(r) = 4 + 8a^2 + \mathcal{O}(r^2).$$

Likewise, insert the far-field decay $f(r) = Cr^{-2} + o(r^{-2})$ and the corresponding decay of f'. Every correction beyond the constant term is of order at most r^{-4}. □

6 No-negative-mode theorem

The monotone hedgehog minimiser already rules out negative spectrum in the hedgehog sector.

Theorem 6.1 (No negative hedgehog mode)**.** *Let f_* be the monotone hedgehog minimiser. Then the bottom of the hedgehog Jacobi spectrum satisfies*

$$\lambda_0 \geq 0.$$

Equivalently, the hedgehog second variation has no negative mode.

Proof. Let $\eta \in C_c^\infty(0, \infty)$. Since $f_*'(r) < 0$ everywhere on the support of η, there exists $\varepsilon_0 > 0$ such that

$$f_* + \varepsilon\eta$$

remains nonincreasing for all $|\varepsilon| < \varepsilon_0$. Because f_* minimises the energy over the monotone class $\mathcal{A}_{\text{mon}}$,

$$E[f_* + \varepsilon\eta] \geq E[f_*] \qquad (|\varepsilon| < \varepsilon_0).$$

Hence the second derivative at $\varepsilon = 0$ is nonnegative:

$$\delta^2 E[f_*](\eta) \geq 0.$$

Density of $C_c^\infty(0, \infty)$ in the natural form domain then gives nonnegativity of the closed quadratic form associated with $\mathcal{J}_{f_*}$, hence $\lambda_0 \geq 0$. □

This theorem is exact, not numerical. It removes the entire negative spectrum. What remains is only the zero-mode obstruction.

7 Reduction of strict positivity to shooting nondegeneracy

7.1 The regular shooting family

Let $f(r;a)$ denote the regular local shooting family

$$f(r;a) = \pi - ar + b(a)r^3 + c(a)r^5 + \mathcal{O}(r^7)$$

with $a > 0$ the origin slope parameter. Let a_* denote the unique parameter corresponding to the monotone decaying branch.

Define

$$\xi(r) := \partial_a f(r;a)\Big|_{a=a_*}. \tag{18}$$

Proposition 7.1. *The function ξ solves the Jacobi equation*

$$\mathcal{J}_{f_*}\xi = 0.$$

Proof. Differentiate the nonlinear profile equation (9) with respect to a along the regular local family. The linearised equation is exactly the Jacobi equation. □

Near the origin,

$$\xi(r) = -r + b'(a_*)r^3 + c'(a_*)r^5 + \mathcal{O}(r^7),$$

so ξ spans the regular origin branch of the Jacobi equation.

7.2 Far-field structure of Jacobi solutions

At infinity, $\mathcal{J}_{f_*}\eta = 0$ reduces to

$$-2(r^2\eta')' + 4\eta = 0,$$

or equivalently

$$r^2\eta'' + 2r\eta' - 2\eta = 0$$

to leading order. Thus every Jacobi solution has asymptotic form

$$\eta(r) = Ar + Br^{-2} + o(r^{-2}), \qquad \eta'(r) = A - 2Br^{-3} + o(r^{-3}). \tag{19}$$

The admissible decaying zero-mode class is characterised by

$$A = 0.$$

Definition 7.2 (Finite-radius transversality functional)**.** For a Jacobi field η, define

$$T_R[\eta] := R\,\eta'(R) + 2\eta(R). \tag{20}$$

If $\eta(r) \sim Br^{-2}$, then $T_R[\eta] \to 0$ as $R \to \infty$. If $\eta(r) \sim Ar + Br^{-2}$ with $A \neq 0$, then

$$T_R[\eta] \sim 3AR.$$

Theorem 7.3 (Analytical reduction of strict hedgehog stability)**.** *The following are equivalent:*

i. $\lambda_0 > 0$ *for the hedgehog Jacobi operator;*

ii. $\ker \mathcal{J}_{f_*} = \{0\}$*;*

iii. the regular Jacobi branch ξ *does not belong to the decaying zero-mode class at infinity.*

Proof. Since $\lambda_0 \geq 0$ is already known, strict positivity is equivalent to triviality of the kernel:

$$\lambda_0 > 0 \iff \ker \mathcal{J}_{f_*} = \{0\}.$$

So it remains only to identify the kernel with the decaying branch obstruction.

The regular origin branch of the Jacobi equation is one-dimensional. The function ξ spans that branch because it is generated by differentiating the one-parameter regular local shooting family. Therefore every regular Jacobi solution is a scalar multiple of ξ.

A nontrivial admissible hedgehog zero mode would have to be regular at the origin and decaying at infinity. Since every regular Jacobi solution is a multiple of ξ, such a mode exists if and only if ξ itself lies in the decaying class. Hence

$$\ker \mathcal{J}_{f_*} \neq \{0\} \iff \xi \text{ is decaying.}$$

Taking contrapositives gives the equivalence of kernel triviality and nondecay of ξ. □

This theorem is the main exact result of the paper. Once it is proved, the entire strict-stability problem is reduced to one nondegeneracy question.

8 Numerical localisation of the monotone decaying branch

The remaining task after Theorem 7.3 is purely numerical in character: determine whether the regular Jacobi branch is decaying.

The monotone decaying shooting branch is localised in the final bracket

$$a_* \in [a_L, a_R] = [2.007528219473,\ 2.007528246180], \tag{21}$$

with width

$$\Delta a = a_R - a_L = 2.670700016693672296 \times 10^{-8}. \tag{22}$$

The branch localisation was obtained by coarse overshoot–undershoot search followed by repeated refinement. At this stage the question is no longer whether the branch exists numerically. It is whether the Jacobi branch associated with that localised profile remains decaying.

8.1 Finite-radius branch diagnostics

For a fixed handoff radius R, define

$$Q_R(a) := R f_r(R; a) + 2 f(R; a). \tag{23}$$

Differentiation with respect to a gives

$$Q_R'(a) = R\,\partial_a f_r(R; a) + 2\,\partial_a f(R; a) = R\,\xi_r(R; a) + 2\,\xi(R; a) = T_R[\xi]. \tag{24}$$

At the final numerical handoff radius $R_0 = 100$, the current proof-chain solve gives

$$Q_{R_0}(a_L) = 3.265283594668382146 \times 10^{-7},$$

$$Q_{R_0}(a_M) = -9.297178664343644025 \times 10^{-7},$$

$$Q_{R_0}(a_R) = -2.088603201513339943 \times 10^{-6}.$$

The sign change is consistent with the branch lying inside the bracket, and the corresponding finite-difference transversality estimate is

$$\frac{Q_{R_0}(a_R) - Q_{R_0}(a_L)}{a_R - a_L} = -9.043065660253793681 \times 10^{1}. \tag{25}$$

This finite-radius diagnostic is strongly nonzero and of the correct sign. The main numerical issue, however, is the actual asymptotic coefficient A, not merely the proxy

(25). We therefore turn directly to asymptotic extraction.

9 Asymptotic coefficient extraction

9.1 Algebraic extraction of A and B

If

$$\xi(r) = Ar + Br^{-2}, \qquad \xi_r(r) = A - 2Br^{-3},$$

then solving for A and B gives

$$A(r) = \frac{2\xi(r) + r\xi_r(r)}{3r}, \tag{26}$$

and

$$B(r) = r^2\Big(\xi(r) - A(r)r\Big). \tag{27}$$

For the exact variable-coefficient Jacobi problem these become radius-dependent estimators, and the question is whether they settle to a nonzero tail limit.

9.2 Large-radius coefficient structure

Writing the Jacobi equation in normalised second-order form

$$\xi'' + P(r)\xi' + Q(r)\xi = 0, \tag{28}$$

the tail proof chain gives

$$P(r) = \frac{2}{r} + \delta P(r), \qquad Q(r) = -\frac{2}{r^2} + \delta Q(r),$$

with conservative bounds

$$p^* := \sup_{r\in[100,400]} |r^2(P(r) - 2/r)| \le 2.235822638141371499 \times 10^{-8}, \tag{29}$$

and

$$q^* := \sup_{r\in[100,400]} |r^4(Q(r) + 2/r^2)| \le 7.448675194438730829 \times 10^{-3}. \tag{30}$$

These are conservative inputs used in the transported asymptotic band.

9.3 Finite-radius input bands at $R_0 = 100$

At the handoff radius $R_0 = 100$, the proof-chain data give

$$A(a_L; R_0) = -2.980409413075540970 \times 10^{-1},$$

$$A(a_M; R_0) = -2.980409486012908782 \times 10^{-1},$$

$$A(a_R; R_0) = -2.980409561416958453 \times 10^{-1},$$

and

$$B(a_L; R_0) = -2.638494066360408397,$$

$$B(a_M; R_0) = -2.638461065558317387,$$

$$B(a_R; R_0) = -2.638426972971785744.$$

Using endpoint hulls, midpoint derivative widening, and explicit conservative padding, the current proof chain takes

$$A(R_0) \in [-2.980409571416958725 \times 10^{-1},\ -2.980409403075540697 \times 10^{-1}], \tag{31}$$

and

$$B(R_0) \in [-2.638504066360408462,\ -2.638416972971785679]. \tag{32}$$

The widening scales used at R_0 are

$$\frac{1}{2} \max |A_a(R_0)|\, \Delta a = 7.556253932079663758 \times 10^{-9}, \tag{33}$$

and

$$\frac{1}{2} \max |B_a(R_0)|\, \Delta a = 3.420062785218068463 \times 10^{-5}. \tag{34}$$

9.4 Exact A/B tail transport

Starting from the conservative finite-radius bands (31) and (32), and transporting them through the exact A/B tail system on $[R_0, R_1]$ with $R_1 = 400$, the current proof chain yields

$$A(R_1) \in [-2.980409918766855903 \times 10^{-1},\ -2.980409055725643519 \times 10^{-1}], \tag{35}$$

and

$$B(R_1) \in [-2.860877386552786561,\ -2.416043652779407580]. \tag{36}$$

The explicit remainder beyond R_1 satisfies

$$|A(\infty) - A(R_1)| \leq 1.850014408356206189 \times 10^{-6}. \tag{37}$$

Therefore the final transported asymptotic band is

$$A(\infty) \in [-2.980428418910939237 \times 10^{-1},\ -2.980390555581560186 \times 10^{-1}], \tag{38}$$

which lies strictly below zero.

Proposition 9.1 (Current asymptotic nondegeneracy closure at the numerical level)**.** *The present proof chain excludes the regular Jacobi branch from the decaying zero-mode class throughout the entire final shooting bracket. In particular, the transported asymptotic band* (38) *implies*

$$A(\infty) < 0 \qquad \textit{for all } a \in [a_L, a_R]$$

at the numerical level now reached.

Proof. The band (38) is entirely negative. Since the decaying class is characterised by $A = 0$, the regular Jacobi branch is numerically excluded from the decaying class on the entire final bracket. □

This is exactly the numerical statement required by Theorem 7.3. It is strong numerical evidence for triviality of the hedgehog kernel and hence strict positivity of the hedgehog Jacobi bottom. It is not yet the final theorem because the input bands and tail coefficients are still produced by floating-point solves and conservative numerical enclosure rather than fully validated interval propagation.

10 What is proved and what remains

It is useful to state the status in one place.

10.1 Analytically proved

The following are analytically established.

1. Existence of a minimiser $f_* \in \mathcal{A}_{\text{mon}}$.
2. Strict monotonicity $f_*'(r) < 0$ for all $r > 0$.
3. Smoothness of f_* on $(0, \infty)$.
4. The regular origin expansion (10) with coefficients (11) and (12).

5. Quadratic far-field decay $f_*(r) = Cr^{-2} + o(r^{-2})$.

6. The exact profile equation (9).

7. The virial identity $E_2[f_*] = E_4[f_*]$.

8. The exact Jacobi operator (17) and endpoint asymptotics of V_{eff}.

9. Non-negativity of the hedgehog Jacobi bottom:

$$\lambda_0 \geq 0.$$

10. The strict-positivity reduction theorem:

$$\lambda_0 > 0 \iff \ker \mathcal{J}_{f_*} = \{0\} \iff \xi \text{ is not decaying.}$$

10.2 Numerically established

The following are numerically established at the current stage.

1. Sharp branch localisation in the final bracket (21).

2. Strong finite-radius transversality diagnostics of the correct sign.

3. Stable asymptotic extraction of the growing coefficient A.

4. Conservative finite-radius input bands for $A(R_0)$ and $B(R_0)$ at $R_0 = 100$.

5. Exact A/B tail transport from R_0 to $R_1 = 400$.

6. A final transported asymptotic band (38) which remains strictly negative across the entire final shooting bracket.

10.3 Remaining certification step

The proof boundary is now isolated to one explicit task: certify the asymptotic exclusion already seen numerically. Concretely, one must replace the current numerical bands for $A(R_0)$, $B(R_0)$, p^*, and q^* by validated interval enclosures on the final bracket, together with a rigorous tail lemma carrying finite-radius enclosure to the true asymptotic coefficient.

Once that is done, Theorem 7.3 upgrades immediately from numerical exclusion to a full strict-positivity theorem.

11 The certification route

The proof closure now has the correct shape. It should not attempt one giant interval box from the singular origin all the way to infinity. The better route is two-stage.

11.1 Finite-radius enclosure

Choose the handoff radius

$$R_0 = 100.$$

Build validated profile enclosures for the regular family at the final bracket endpoints a_L, a_R, and validated enclosures for the corresponding first sensitivity data

$$\xi(R_0), \qquad \xi_r(R_0).$$

From these, form validated enclosures for

$$A(R_0) = \frac{2\xi(R_0) + R_0\xi_r(R_0)}{3R_0}, \qquad B(R_0) = R_0^2\Big(\xi(R_0) - A(R_0)R_0\Big).$$

11.2 Tail lemma

Using the large-radius form

$$\xi'' + P(r)\xi' + Q(r)\xi = 0, \qquad P(r) = \frac{2}{r} + \delta P(r), \qquad Q(r) = -\frac{2}{r^2} + \delta Q(r),$$

prove a rigorous estimate of the form

$$|A(\infty) - A(R_1)| \le \varepsilon_{\text{tail}}$$

with explicit $\varepsilon_{\text{tail}}$ derived from interval bounds on δP and δQ, or equivalently on p^* and q^*.

11.3 Bracket widening

Combine the finite-radius enclosure with a validated bound on A_a over the final bracket:

$$|A(a) - A(a_M)| \le \sup |A_a| \, \frac{\Delta a}{2}.$$

Since the current numerical margin is of order 10^{-1} while the bracket-width correction is of order 10^{-8}, the required rigorous widening is not tight. It only needs to be honest.

11.4 Conclusion of the strict-stability theorem

If the final validated band satisfies

$$0 \notin A(\infty)\Big([a_L, a_R]\Big),$$

then the regular Jacobi branch is rigorously excluded from the decaying class. By Theorem 7.3,

$$\ker \mathcal{J}_{f_*} = \{0\}, \qquad \lambda_0 > 0.$$

That is the exact remaining proof task.

12 Conclusion

The strict hedgehog-sector stability problem for the frozen DTFE matter branch has now been reduced to the correct scale.

The analytical side is already clean. In the monotone hedgehog class there exists a finite-energy minimiser, it is strictly decreasing and smooth, its profile obeys the exact frozen DTFE hedgehog equation, its scaling balance gives the virial identity, the hedgehog second variation has no negative mode, and strict positivity is equivalent to exclusion of the regular Jacobi branch from the decaying zero-mode class.

The numerical side is now also clean in the correct sense. The final shooting window is sharply localised, the finite-radius transversality diagnostics are strongly nonzero, the finite-radius input bands at $R_0 = 100$ are stable, and exact A/B tail transport gives a bracket-wide asymptotic band

$$A(\infty) \in [-2.980428418910939237 \times 10^{-1},\ -2.980390555581560186 \times 10^{-1}]$$

that remains strictly negative throughout the final bracket.

So the problem has changed character. The analytical structure is complete, and the numerical nondegeneracy target is resolved in the correct asymptotic variable. What remains is not further exploration but formal certification. Once the present bracket-wide asymptotic exclusion is enclosed by validated interval methods, strict hedgehog-sector stability follows immediately from the analytical reduction theorem already proved.

A Regular local expansion

For convenience we record the local regular branch again:

$$f(r; a) = \pi - ar + b(a)r^3 + c(a)r^5 + \mathcal{O}(r^7),$$

with

$$b(a) = \frac{a^3(a^2+2)}{15(2a^2+1)},$$

and

$$c(a) = \frac{a^5\Big(92a^8 + 316a^6 + 459a^4 + 330a^2 + 120\Big)}{3150(2a^2+1)^3(5a^2+3)}.$$

Its a-derivative satisfies

$$\xi(r;a) = -r + b'(a)r^3 + c'(a)r^5 + \mathcal{O}(r^7),$$

with

$$b'(a) = \frac{a^2(2a^4+3a^2+2)}{5(2a^2+1)^2}.$$

B Numerical summary tables

Table 1 records the midpoint tail values for the asymptotic coefficients $A_{\mathrm{mid}}(r)$ and $B_{\mathrm{mid}}(r)$. Table 2 records the corresponding derivative data $A'_{\mathrm{mid}}(r)$ and $A_{a,\mathrm{mid}}(r)$.

Table 1: Representative midpoint tail values for $A_{\mathrm{mid}}(r)$ and $B_{\mathrm{mid}}(r)$.

r	$A_{\mathrm{mid}}(r)$	$B_{\mathrm{mid}}(r)$
100	$-2.980409486012908782 \times 10^{-1}$	-2.638461065558317387
150	$-2.980409452900610878 \times 10^{-1}$	-2.644607211781391243
200	$-2.980409446728651823 \times 10^{-1}$	-2.647651699874131737
300	$-2.980409444489772186 \times 10^{-1}$	-2.650613277666025169
400	$-2.980409444140529329 \times 10^{-1}$	-2.651982545103237499

Table 2: Representative midpoint tail values for $A'_{\mathrm{mid}}(r)$ and $A_{a,\mathrm{mid}}(r)$.

r	$A'_{\mathrm{mid}}(r)$	$A_{a,\mathrm{mid}}(r)$
100	$1.849126517415104437 \times 10^{-10}$	$-5.658586328120769338 \times 10^{-1}$
150	$2.418316812716721317 \times 10^{-11}$	$-5.675591724826708484 \times 10^{-1}$
200	$5.664127659376395444 \times 10^{-12}$	$-5.684055432026805521 \times 10^{-1}$
300	$7.064047336085540593 \times 10^{-13}$	$-5.692403176751145155 \times 10^{-1}$
400	$1.501326912905214558 \times 10^{-13}$	$-5.696416499705965197 \times 10^{-1}$

The tail is evidently settled. The remaining challenge is not signal strength but rigorous enclosure.

C Proof-chain summary

For convenience we record the final proof-chain values used in the transported asymptotic band.

$$
\begin{aligned}
A(R_0) &\in [-2.980409571416958725 \times 10^{-1},\ -2.980409403075540697 \times 10^{-1}],\\
B(R_0) &\in [-2.638504066360408462,\ -2.638416972971785679],\\
p^* &\le 2.235822638141371499 \times 10^{-8},\\
q^* &\le 7.448675194438730829 \times 10^{-3},\\
A(\infty) &\in [-2.980428418910939237 \times 10^{-1},\ -2.980390555581560186 \times 10^{-1}].
\end{aligned}
$$

These values are conservative numerical inputs to the current proof chain. They are not yet validated interval certificates.

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